

Tema 3

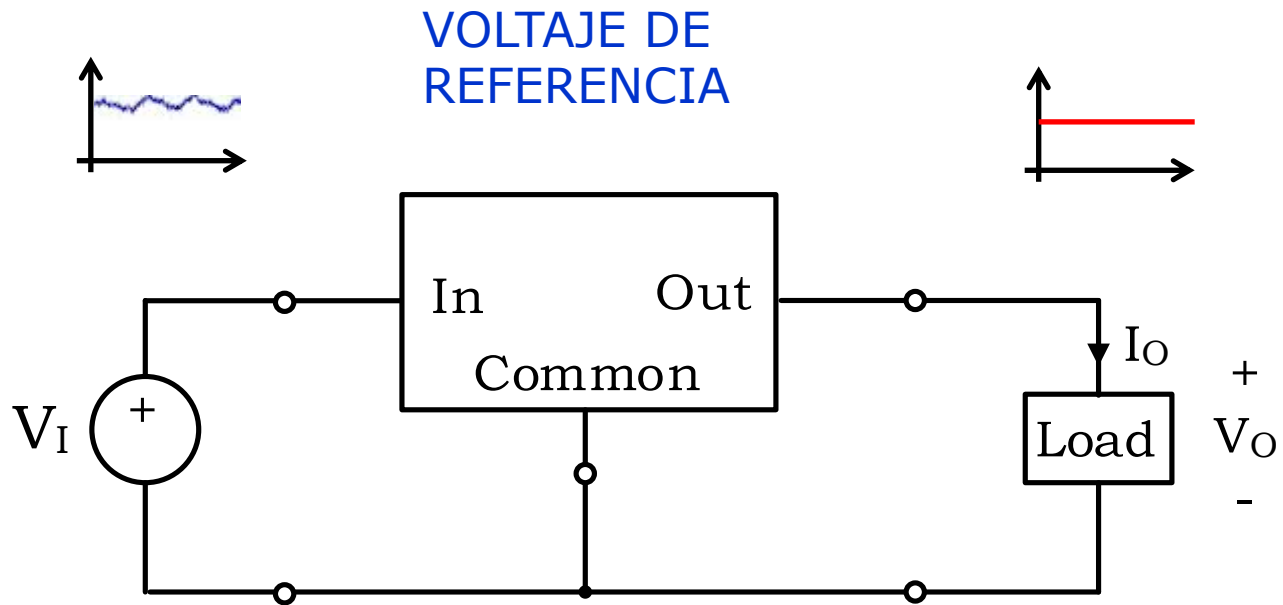


Tensiones de Referencia
(V_{REF})

Índice

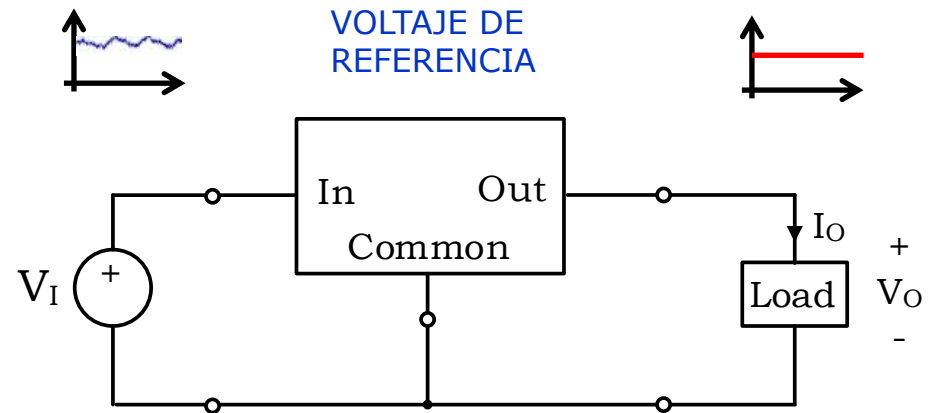
1. Funcionalidad
2. Especificaciones
3. Fundamentos
4. Ejemplos

Funcionalidad



Proporciona un **voltaje estable** independiente de las condiciones de trabajo (V_I , I_o , $T...$)

Funcionalidad



- Proporcionan una tensión patrón.
- Tienen alta precisión y estabilidad.
- Proporcionan corrientes de salida muy bajas.
- La precisión de multímetros, ADC's, DAC's, convertidores V-F y F-V, circuitos VCO's, etc. Dependen de la calidad de la tensión de referencia que empleen.
- No confundir con los reguladores de tensión, con una función similar, pero de menor calidad y válidos para alimentar otros circuitos.

Índice

1. Funcionalidad
- 2. Especificaciones**
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Especificaciones principales

❑ Regulación de línea

$$R.L. = \frac{\Delta V_0}{\Delta V_I} \quad [\text{mV/V}]$$

$$R.L.(\%) = \frac{\Delta V_0}{\Delta V_I} \frac{1}{V_O} \cdot 100 \quad [\%/V]$$

❑ Regulación de carga

$$R.C. = \frac{\Delta V_0}{\Delta I_0} \quad [\text{mV/mA}]$$

$$R.C.(\%) = \frac{\Delta V_0}{\Delta I_0} \frac{1}{V_o} \cdot 100 \quad [\%/mA]$$

Especificaciones principales

❑ Coeficiente de temperatura

$$TC(V_0) = \frac{\Delta V_0}{\Delta T} \quad [V/^{\circ}C]$$

$$TC(V_0) = \frac{\Delta V_0}{\Delta T} \cdot \frac{1}{V_0} \cdot 100 \quad [^{\circ}/^{\circ}C]$$

$$TC(V_0) = \frac{\Delta V_0}{\Delta T} \cdot \frac{1}{V_0} \cdot 10^6 \quad [ppm/^{\circ}C]$$

❑ Ruido

Por ejemplo $6 \mu V_{pp}$ desde 0.1 Hz a 10 Hz.

❑ Long term stability

Por ejemplo 50 [ppm/(1000 h)]

Para $V_{ref} = 10 V$, en unos 42 días
 V_{ref} ha podido cambiar en 0,5 mV

❑ Dropout Voltage

Diferencia de tensión necesaria mínima, para buen funcionamiento, entre la tensión de entrada y la de salida de la referencia de tensión

Ejercicio Se dispone de la referencia de tensión REF101KM (Burr-Brown) con los siguientes parámetros medidos a: $T_A = 25\text{ °C}$ y $V_I = 15\text{ V}$

$$V_{REF} = 10\text{ V} \quad RL = 0.001 \left[\frac{\%}{V} \right] \quad RC = 0.001 \left[\frac{\%}{mA} \right] \quad TC = 1 \left[\frac{ppm}{\text{°C}} \right]$$

Calcular la variación máxima de tensión en su salida debido a:

a.- Variación en su tensión de entrada: $13.5\text{ V} \leq V_I \leq 35\text{ V}$

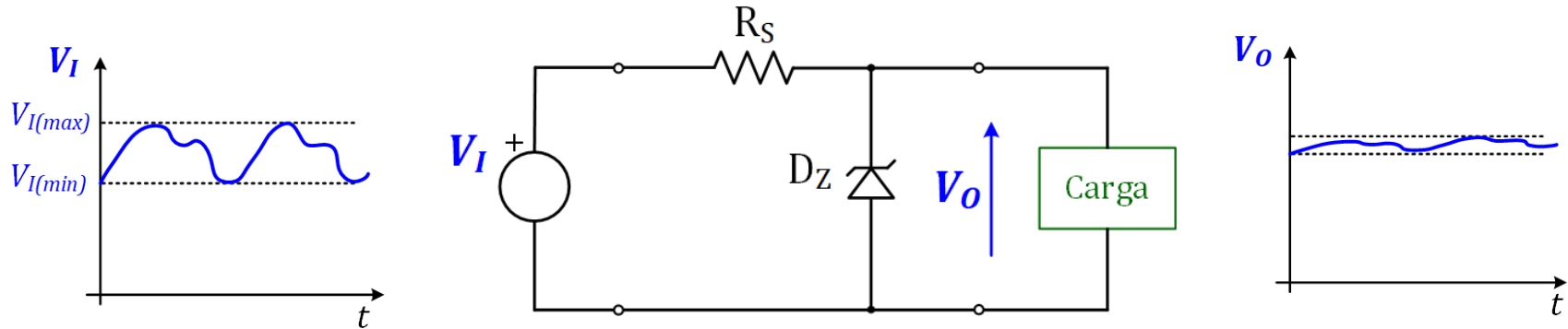
b.- Variación en su corriente de salida: $\Delta I_o = \pm 10\text{ mA}$

c.- Temperatura de trabajo: $0\text{ °C} \leq T \leq 70\text{ °C}$

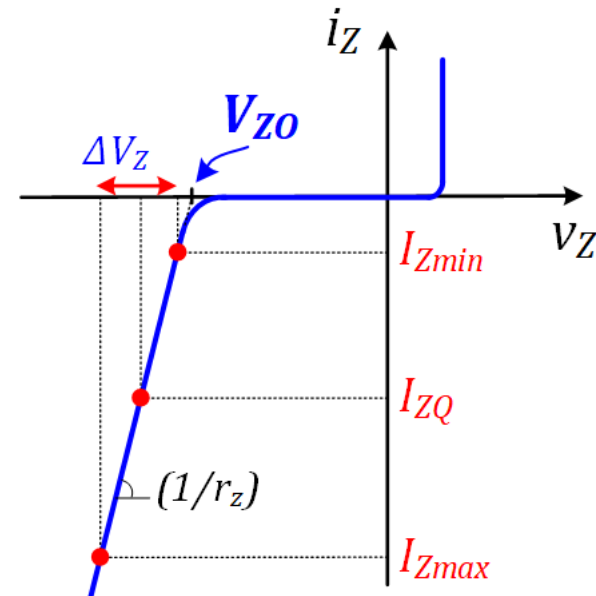
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1ª aprox. a V_{REF} : Regulador con Zener



La clave está en que la tensión Zener cambia muy poco aunque cambie mucho la corriente Zener.

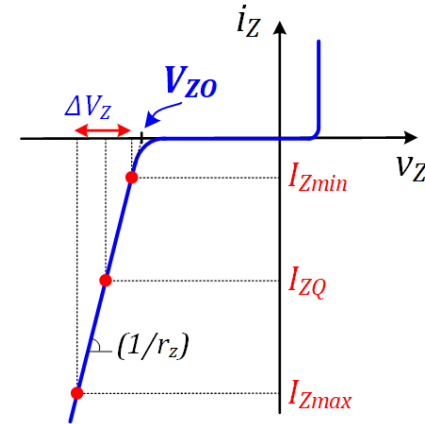


Diseño y caracterización de un Regulador con Zener

Empleamos un Zener con las siguientes características, medidas a potencia mitad de la máxima:

$$V_Z = 6.8 \text{ V} \quad P_{max} = 0.5 \text{ W} \quad r_z = 10 \Omega$$

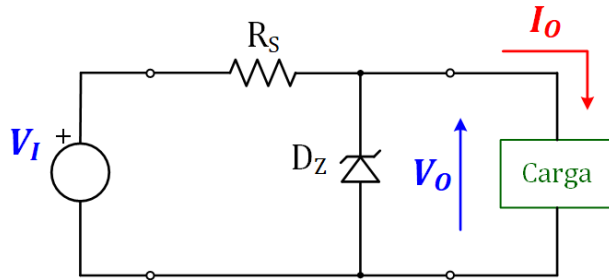
¿Cuál es el valor de V_{ZO} , I_{Zmax} e I_{Zmin} ?



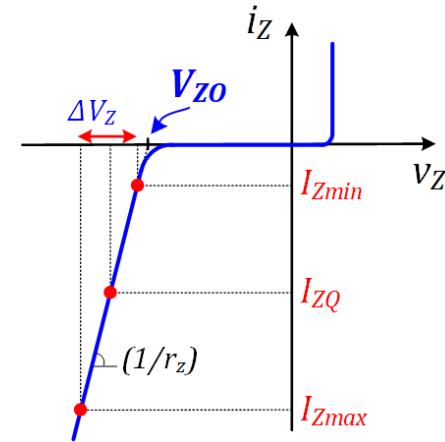
Diseño y caracterización de un Regulador con Zener

El regulador deberá trabajar con:

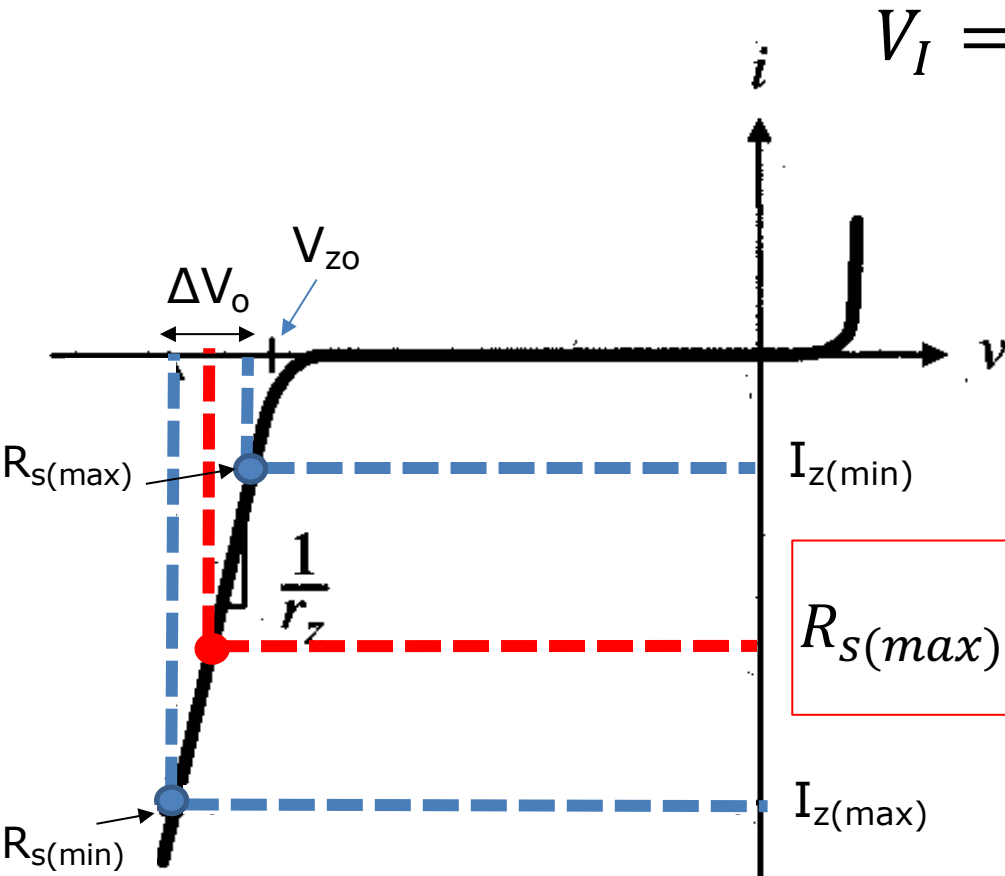
$$10V \leq V_I \leq 20V \quad 0 \leq I_O \leq 10mA$$



Proponga un valor adecuado para R_S



Cálculo de R_s



$$V_I = R_S \cdot (I_Z + I_o) + r_z \cdot I_Z + V_{z_o}$$

$$R_s = \frac{V_I - (r_z \cdot I_z + V_{z_o})}{I_z + I_o}$$

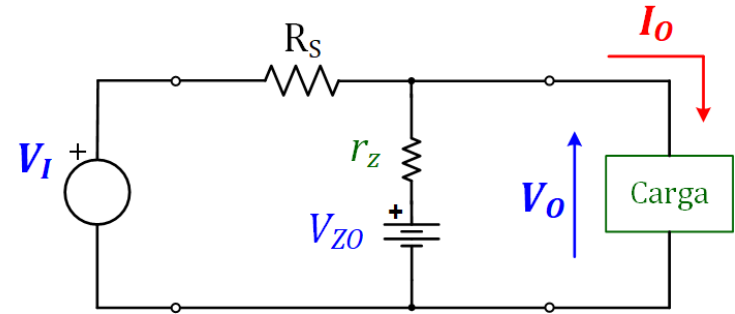
$$R_{s(max)} = \frac{V_{I(min)} - (r_z \cdot I_{z(min)} + V_{z_o})}{I_{z(min)} + I_{o(max)}}$$

$$R_{S(min)} = \frac{V_{I(max)} - (r_z \cdot I_{z(max)} + V_{z_o})}{I_{z(max)} + I_{o(min)}}$$

Diseño y caracterización de un Regulador con Zener

Teniendo en cuenta el modelo lineal del Zener:

Obtenga la expresión que relaciona V_o con V_I e I_o .



Obtenga los parámetros RL y RC del regulador.

Circuito Autoregulado

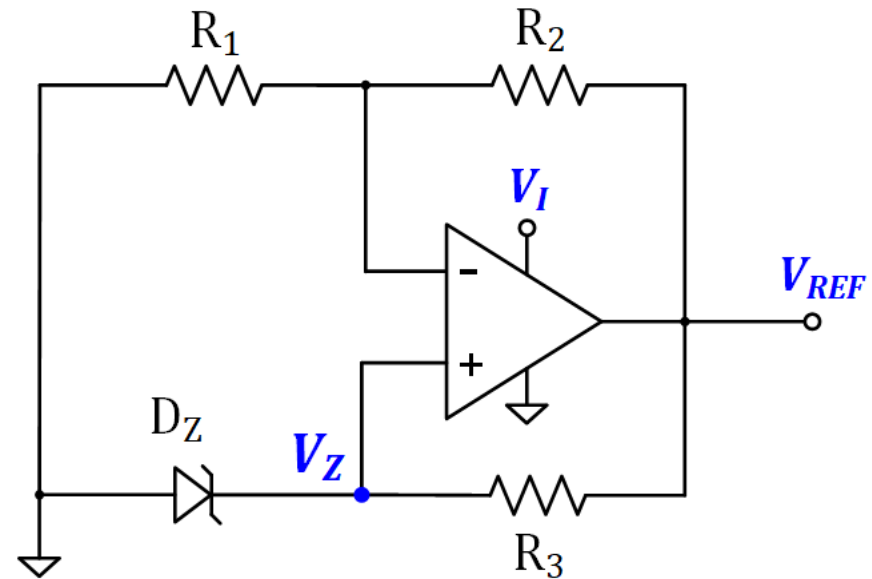
Emplea un Zener, un AO y tres resistores. La tensión de alimentación del AO es la tensión de entrada V_I .

- El Zener se polariza mediante la propia referencia de tensión:

$$I_{ZQ} = \frac{(V_{REF} - V_Z)}{R_3}$$

- El valor de la referencia de tensión es:

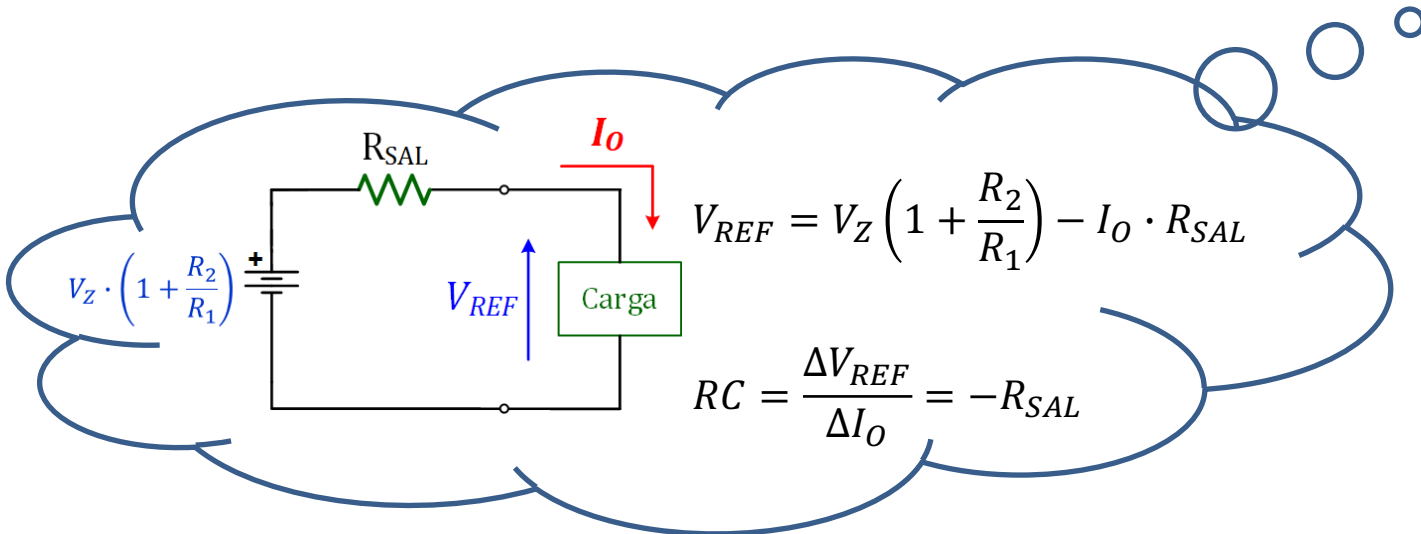
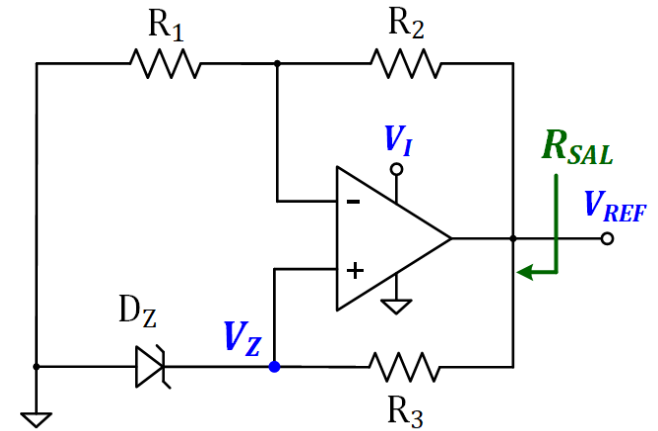
$$V_{REF} = V_Z \cdot \left(1 + \frac{R_2}{R_1}\right)$$



Circuito Autoregulado

Mejora los parámetros RL y RC del Zener gracias al AO.

- Parámetro Regulación de Carga, RC:



$$RC = \frac{\Delta V_{REF}}{\Delta I_O} = -R_{SAL} \cong -\frac{r_o}{1 + a\beta} \quad \text{con} \quad \beta \cong \frac{1}{A_{V(ideal)}} = \frac{R_1}{R_1 + R_2}$$

Circuito Autoregulado

Mejora los parámetros RL y RC del Zener gracias al AO.

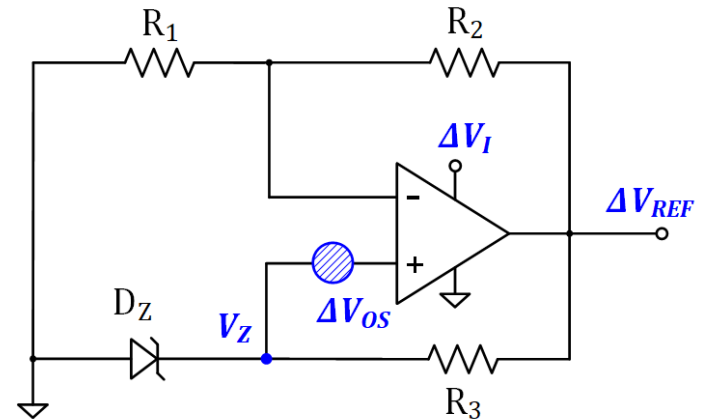
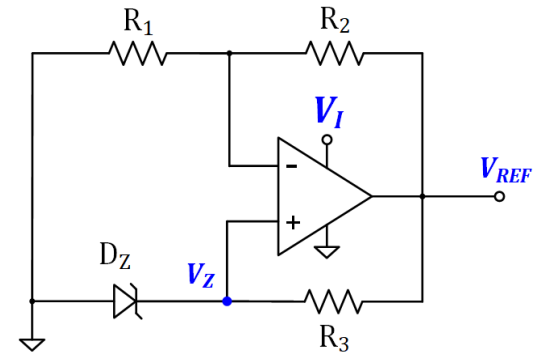
- Parámetro Regulación de Línea, RL:

$$RL = \frac{\Delta V_{REF}}{\Delta V_I}$$

Cuando cambia V_I cambia la tensión offset y la tensión en modo común a la entrada del AO.

$$\Delta V_{OS} = \frac{\Delta V_I}{PSR} + \frac{\overbrace{0.5 \cdot \Delta V_I}^{\Delta V_{CM}}}{CMR}$$

$$\begin{aligned} \Delta V_{REF} &= \Delta V_{OS} \left(1 + \frac{R_2}{R_1} \right) = \\ &= \left(1 + \frac{R_2}{R_1} \right) \cdot \Delta V_I \cdot \left(\frac{1}{PSR} + \frac{0.5}{CMR} \right) \end{aligned}$$



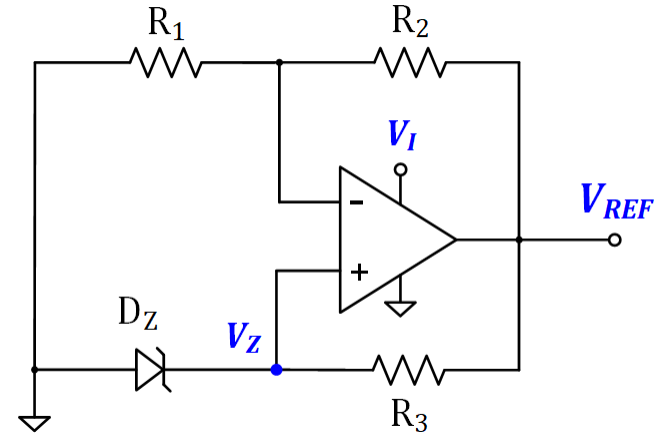
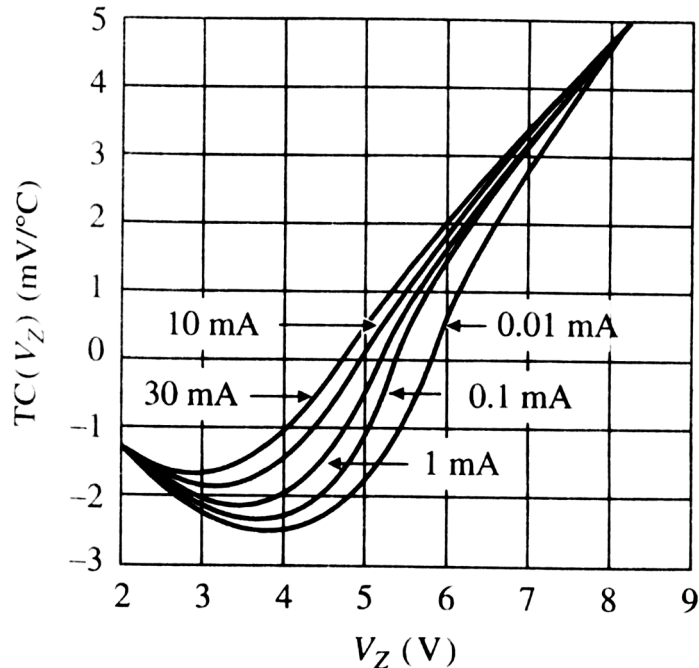
$$RL = \frac{\Delta V_{REF}}{\Delta V_I} = \left(1 + \frac{R_2}{R_1} \right) \left(\frac{1}{PSR} + \frac{0.5}{CMR} \right)$$

PSR y CMR en unidades naturales

Circuito Autoregulado

La estabilidad térmica de V_{REF} no puede ser mejor que la de V_Z .

El TC de un Zener depende de V_Z y de I_Z .

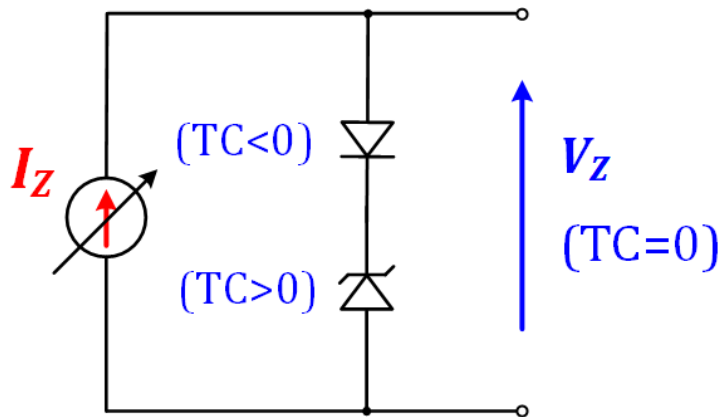
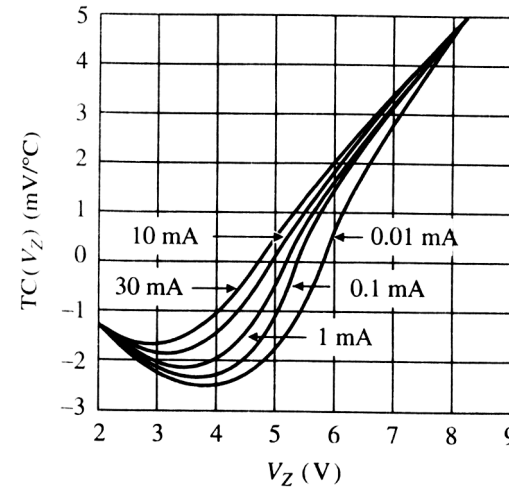


Como se observa TC, puede ser positivo o negativo, dependiendo del valor del Zener y de la corriente que circule a su través.

Circuito Autoregulado: Compensacion del TC

El TC de un Zener depende de V_Z y de I_Z .

Conectando en serie con el Zener ($TC > 0$) un diodo ($TC < 0$) se puede conseguir $TC \cong 0$. Hay que ajustar la I_Z .



$$TC(V_D) \cong -2.1 [mV/^\circ C]$$

Coeficiente de temperatura de un diodo

Unión PN polarizada en directo.

$$TC(V_D) = - \left(\frac{V_{GO} - V_D}{T} + \frac{3k}{q} \right)$$

V_{GO} = Tensión Bandgap para el Silicio = 1,205 V.

T = Temperatura Absoluta.

k = Constante de Boltzman: $1,381 \times 10^{-23}$

q = Carga del electrón: $1,6 \times 10^{-19}$

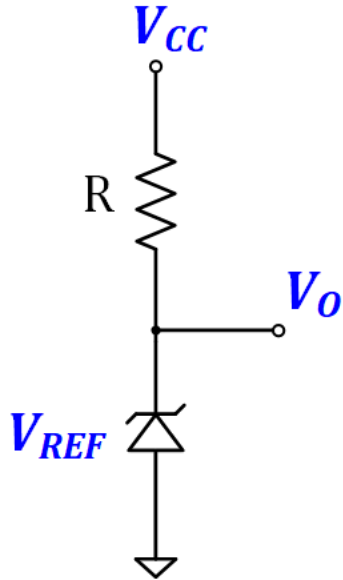
$$TC(V_D) \approx -2mV/^{\circ}C$$

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- 4. Ejemplos**

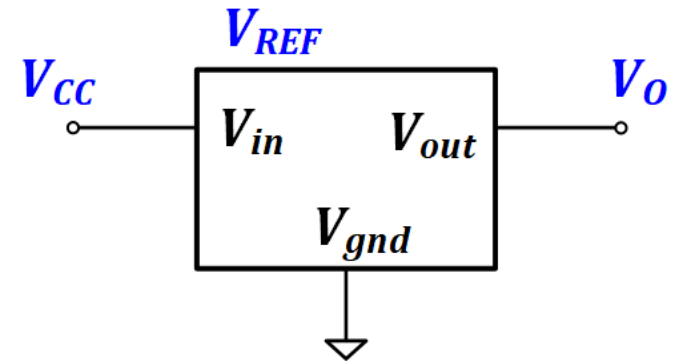
Modos de Operación

Modo Shunt



- Menos preciso
- Tensiones más altas de operación
- Mayor corriente de operación

Modo Serie



LM136-2.5/LM236-2.5/LM336-2.5V Reference Diode

General Description

The LM136-2.5/LM236-2.5 and LM336-2.5 integrated circuits are precision 2.5V shunt regulator diodes. These monolithic IC voltage references operate as a low-temperature-coefficient 2.5V zener with 0.2Ω dynamic impedance. A third terminal on the LM136-2.5 allows the reference voltage and temperature coefficient to be trimmed easily.

The LM136-2.5 series is useful as a precision 2.5V low voltage reference for digital voltmeters, power supplies or op amp circuitry. The 2.5V make it convenient to obtain a stable reference from 5V logic supplies. Further, since the LM136-2.5 operates as a shunt regulator, it can be used as either a positive or negative voltage reference.

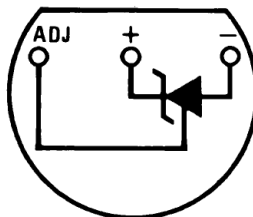
The LM136-2.5 is rated for operation over -55°C to $+125^{\circ}\text{C}$ while the LM236-2.5 is rated over a -25°C to $+85^{\circ}\text{C}$ temperature range.

The LM336-2.5 is rated for operation over a 0°C to $+70^{\circ}\text{C}$ temperature range. See the connection diagrams for available packages.

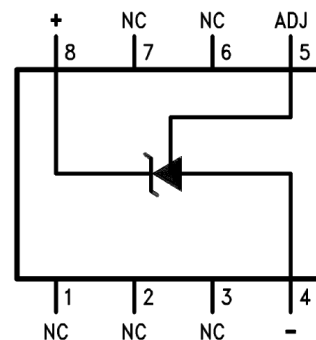
Features

- Low temperature coefficient
- Wide operating current of $400\mu\text{A}$ to 10 mA
- 0.2Ω dynamic impedance
- $\pm 1\%$ initial tolerance available
- Guaranteed temperature stability
- Easily trimmed for minimum temperature drift
- Fast turn-on

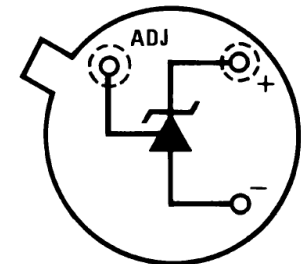
TO-92
Plastic Package



SO Package



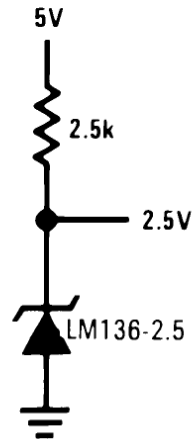
TO-46
Metal Can Package



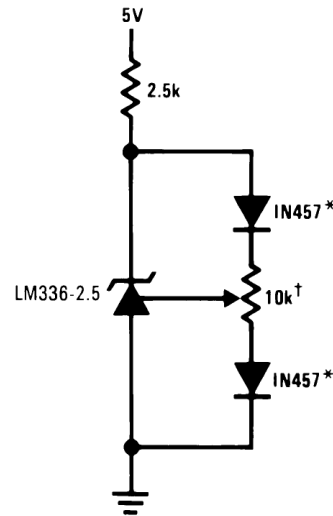
LM136-2.5/LM236-2.5/LM336-2.5V Reference Diode

Typical Applications

2.5V Reference



**2.5V Reference with Minimum
Temperature Coefficient**

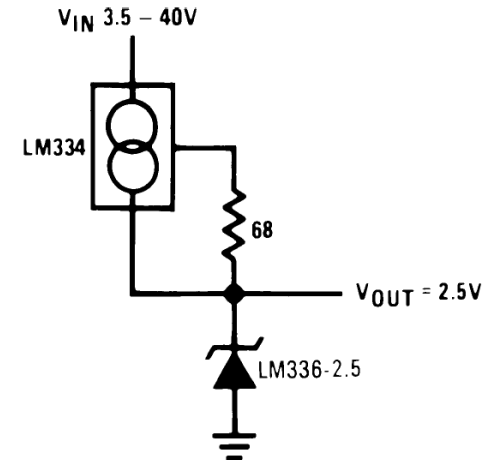


†Adjust to 2.490V

*Any silicon signal diode

00571510

Wide Input Range Reference



LM334 es una fuente de corriente ajustable

LM136-2.5/LM236-2.5/LM336-2.5V Reference Diode

Electrical Characteristics

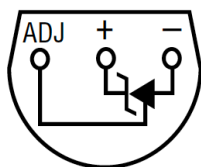
Parameter	Conditions	LM336B-2.5 LM336-2.5			Units
		Min	Typ	Max	
Reverse Breakdown Voltage	$T_A = 25^\circ\text{C}$, $I_R = 1\text{ mA}$ LM136, LM236, LM336 LM136A, LM236A, LM336B	2.390 2.440	2.490 2.490	2.590 2.540	V V
Reverse Breakdown Change With Current	$T_A = 25^\circ\text{C}$, $400\text{ }\mu\text{A} \leq I_R \leq 10\text{ mA}$		2.6	10	mV
Reverse Dynamic Impedance	$T_A = 25^\circ\text{C}$, $I_R = 1\text{ mA}$, $f = 100\text{ Hz}$		0.2	1	Ω
Temperature Stability (Note 4)	V_R Adjusted to 2.490V $I_R = 1\text{ mA}$, (Figure 2) $0^\circ\text{C} \leq T_A \leq 70^\circ\text{C}$ (LM336) $-25^\circ\text{C} \leq T_A \leq +85^\circ\text{C}$ (LM236H, LM236Z) $-25^\circ\text{C} \leq T_A \leq +85^\circ\text{C}$ (LM236M) $-55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$ (LM136)		1.8	6	mV mV mV mV
Reverse Breakdown Change With Current	$400\text{ }\mu\text{A} \leq I_R \leq 10\text{ mA}$		3	12	mV
Reverse Dynamic Impedance	$I_R = 1\text{ mA}$		0.4	1.4	Ω
Long Term Stability	$T_A = 25^\circ\text{C} \pm 0.1^\circ\text{C}$, $I_R = 1\text{ mA}$, $t = 1000\text{ hrs}$		20		ppm

FEATURES

- *Guaranteed* Temperature Stability
- Maximum 0.6Ω Dynamic Impedance
- Adjustable for Minimum Temperature Coefficient
- Wide Operating Current Range

PACKAGE/ORDER INFORMATION

BOTTOM VIEW

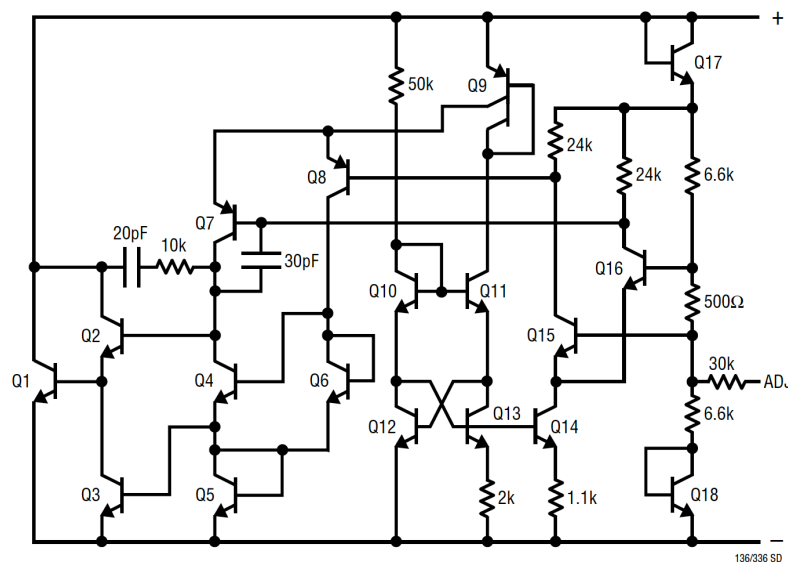

 Z PACKAGE
 3-LEAD PLASTIC TO-92

 $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 160^{\circ}\text{C/W}$
 (NOTE 2)

APPLICATIONS

- Reference for 5V Systems
- 8-Bit A/D and D/A Reference
- Digital Voltmeters
- Current Loop Measurement and Control Systems
- Power Supply Monitor

SCHEMATIC DIAGRAM



ELECTRICAL CHARACTERISTICS

The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^\circ\text{C}$. (Note 2)

SYMBOL	PARAMETER	CONDITIONS	LM136/LM136A			LM336/LM336B			UNITS
			MIN	TYP	MAX	MIN	TYP	MAX	
V_Z	Reverse Breakdown Voltage	$T_A = 25^\circ\text{C}$, $I_R = 1\text{mA}$ LM136/LM336 LM136A/LM336B	2.440	2.490	2.540	2.390	2.490	2.590	V
			2.465	2.490	2.515	2.440	2.490	2.540	V
$\frac{\Delta V_Z}{\Delta I_R}$	Reverse Breakdown Change with Current	$400\mu\text{A} \leq I_R \leq 10\text{mA}$ $T_A = 25^\circ\text{C}$ $T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$		2.6	6		2.6	10	mV
				3.0	10		3.0	12	mV
r_Z	Reverse Dynamic Impedance	$I_R = 1\text{mA}$ $T_A = 25^\circ\text{C}$ $T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$		0.2	0.6		0.2	1.0	Ω
				0.4	1.0		0.4	1.4	Ω
$\frac{\Delta V_Z}{\Delta \text{Temp}}$	Temperature Stability	V_R Adjusted to 2.490V, $I_R = 1\text{mA}$, $T_{\text{MIN}} \leq T_A \leq T_{\text{MAX}}$ (See Figure 1)		12	18		1.8	6	mV
$\frac{\Delta V_Z}{\Delta \text{Time}}$	Long Term Stability	$T_A = 25^\circ\text{C} \pm 0.1^\circ\text{C}$, $I_R = 1\text{mA}$		20			20		ppm/kHr

Note 1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

Note 2: The LT1009 is an improved, low cost, pin for pin replacement for the "A" and "B" versions. For further information consult the LT1009 data sheet.

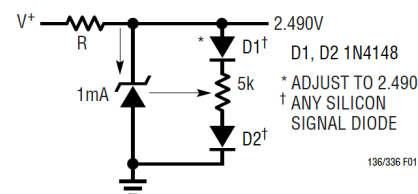
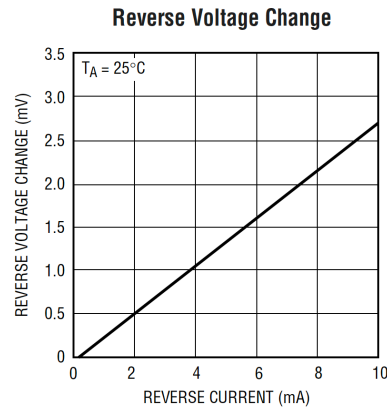
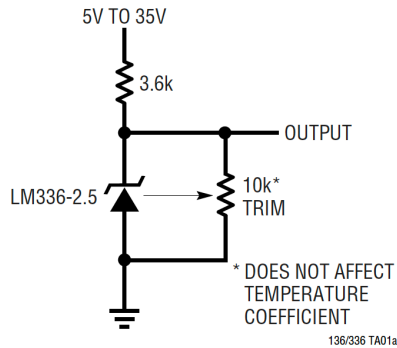


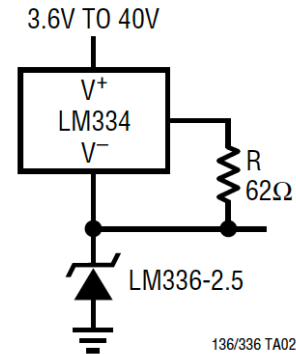
Figure 1. Adjusting the LM336 for Minimum Temperature Coefficient

TYPICAL APPLICATION

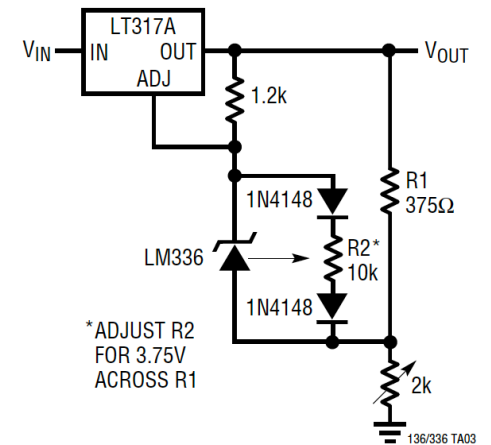
2.5V Reference



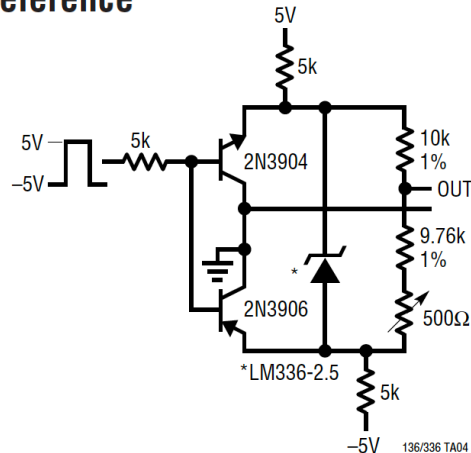
Wide Supply Range, Reference



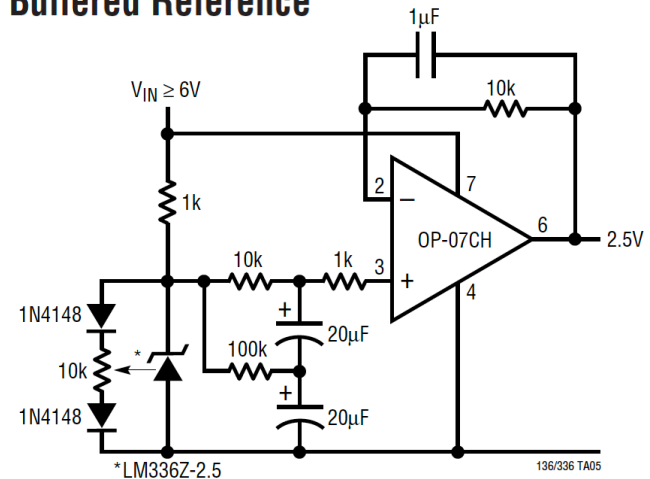
Low Temperature Coefficient Power Regulator



Switchable $\pm 1.25\text{V}$ Bipolar Reference



Low Noise 2.5V Buffered Reference



FEATURES

High output accuracy

REF01: 10.0 V, $\pm 0.3\%$ maximum

REF02: 5.0 V, $\pm 0.3\%$ maximum

REF03: 2.5 V, $\pm 0.6\%$ maximum

Adjustable output: $\pm 3\%$ minimum

Excellent temperature stability

REF01: 8.5 ppm/ $^{\circ}\text{C}$ maximum

REF02: 8.5 ppm/ $^{\circ}\text{C}$ maximum

REF03: 50 ppm/ $^{\circ}\text{C}$ maximum

Low noise

REF01: 30 μV p-p typical

REF02: 15 μV p-p typical

REF03: 6 μV p-p typical

High supply voltage range: up to 36 V maximum

Low supply current: 1.4 mA maximum

High load-driving capability: 10 mA maximum

Temperature output function

APPLICATIONS

Precision data systems

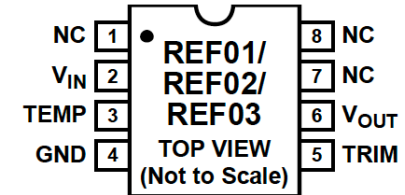
High resolution converters

Industrial process control systems

Precision instruments

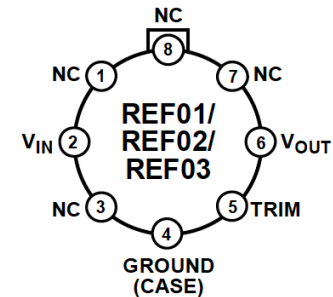
Military and aerospace applications

PIN CONFIGURATIONS



NC = NO CONNECT. DO NOT CONNECT ANYTHING ON THESE PINS. SOME OF THEM ARE RESERVED FOR FACTORY TESTING PURPOSES.

00375-001



NC = NO CONNECT. DO NOT CONNECT ANYTHING ON THESE PINS. SOME OF THEM ARE RESERVED FOR FACTORY TESTING PURPOSES.

00375-002

Table 1. Selection Guide

Device Number	Output Voltage	Input Voltage Range
REF01	10.0 V	12 V to 36 V
REF02	5.0 V	7.0 V to 36 V
REF03	2.5 V	4.5 V to 36 V

APPLICATIONS INFORMATION

BASIC REFERENCE APPLICATION

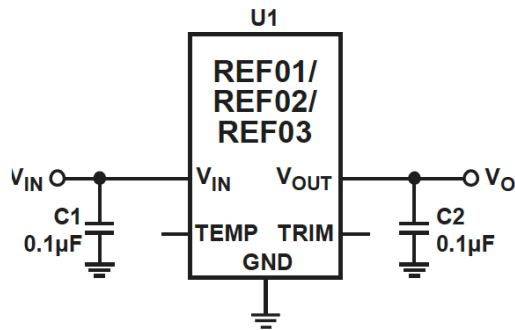


Figure 38. Basic Reference Application

ADJUSTABLE REFERENCE WITH POSITIVE AND NEGATIVE SWING

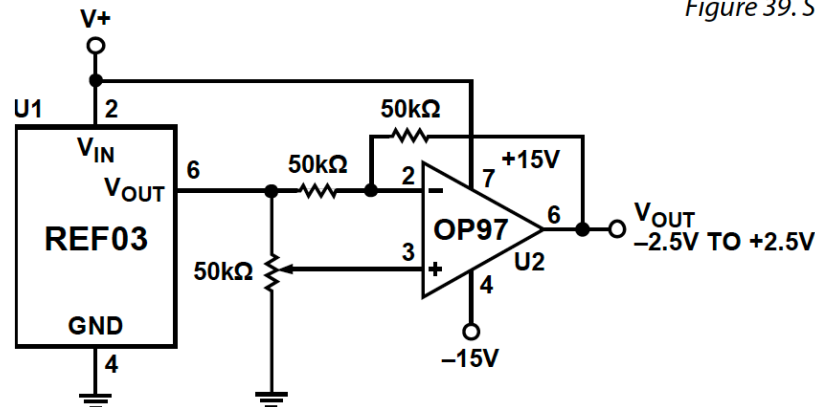


Figure 43. Negatively Adjustable Reference

LOW COST CURRENT SOURCE

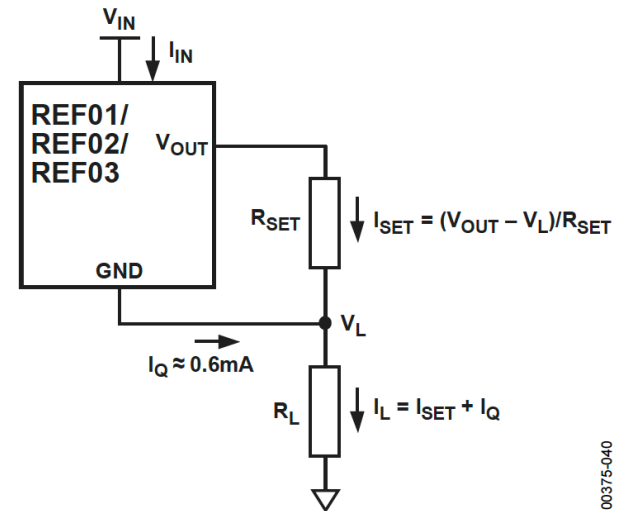


Figure 39. Simple Current Source

REF03 SPECIFICATIONS

$V_{IN} = 15\text{ V}$, $-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$, $I_{LOAD} = 0\text{ mA}$, unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
OUTPUT VOLTAGE	V_O		2.495	2.500	2.515	V
OUTPUT ADJUSTMENT RANGE ¹	ΔV_{TRIM}	POT = 10 k Ω	± 6	± 11		%
INITIAL ACCURACY	V_{OERR}				± 15 ± 0.6	mV %
TEMPERATURE COEFFICIENT	TCV_O			10	50	ppm/ $^{\circ}\text{C}$
LINE REGULATION ²	$\Delta V_O / \Delta V_{IN}$	$V_{IN} = 4.5\text{ V to } 33\text{ V}$		20	50	ppm/V
LOAD REGULATION ²	$\Delta V_O / \Delta I_{LOAD}$	$I_{LOAD} = 0\text{ mA to } 10\text{ mA}$		60	100	ppm/mA
DROPOUT VOLTAGE	V_{DO}				2	V
QUIESCENT CURRENT	I_{IN}			1.0	1.4	mA
LOAD CURRENT	I_{LOAD}					
Sourcing					10	mA
Sinking					-0.3	mA
SHORT CIRCUIT TO GND	I_{SC}	$V_O = 0\text{ V}$		24		mA
VOLTAGE NOISE	$e_{N\text{ p-p}}$	0.1 Hz to 10.0 Hz		6		$\mu\text{V p-p}$
LONG-TERM STABILITY ³	ΔV_O	After 1000 hours of operation		50		ppm
TURN-ON SETTLING TIME	t_R	Output settling to within $\pm 0.1\%$ of final value		5		μs
TEMPERATURE SENSOR ⁴						
Voltage Output at TEMP Pin	V_{TEMP}			580		mV
Temperature Sensitivity	TCV_{TEMP}			1.96		mV/ $^{\circ}\text{C}$

¹ Refer to the Output Adjustment section.

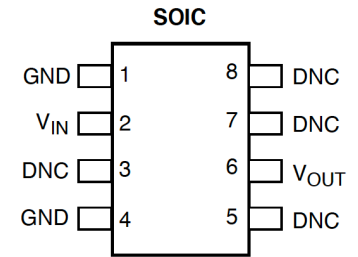
² Specification includes the effects of self-heating.

³ Long-term stability is noncumulative; the drift in subsequent 1000-hour periods is significantly lower than in the first 1000-hour periods. Refer to the [AN-713 Application Note](#).

⁴ Refer to the Temperature Monitoring section.

Precision 5.0V FGA™ Voltage Reference

X60008C-50
X60008D-50



FEATURES

- **Output Voltage: 5.000V**
- **Absolute Initial Accuracy Options:**
±0.5mV & ±1.0mV
- **Ultra Low Power Supply Current: 500nA**
- **Low Temperature Coefficient options:**
5 & 10ppm/°C
- **10 mA Source & Sink Current Capability**
- **10 ppm/1000hrs Long Term Stability**
- **Very Low Dropout Voltage: 100 mV @ no load**
- **Supply Voltage Range: 5.1V to 9.0V**
- **5kV ESD (Human Body Model)**
- **Standard Package: SOIC-8**
- **Temp Range: -40°C to +85°C**

DESCRIPTION

The X60008-50 FGA™ voltage references are very high precision analog voltage references fabricated in Xicor's proprietary Floating Gate Analog technology, which achieves superior levels of performance when compared to conventional band gap, buried zener, or X_{FET}™ technologies.

FGA™ voltage references feature very high initial accuracy, very low temperature coefficient, excellent long term stability, low noise and excellent line and load regulation, at the lowest power consumption currently available. These voltage references enable advanced applications for precision industrial & portable systems operating at significantly higher accuracy and lower power levels than can be achieved with conventional technologies.

APPLICATIONS

- **High Resolution A/Ds & D/As**
- **Digital Meters**
- **Calibration Systems**
- **V-F Converters**
- **Precision Current Sources**
- **Precision Regulators**
- **Precision Oscillators**
- **Battery Management Systems**
- **Smart sensors**
- **Strain Gage Bridges**
- **Threshold Detectors**
- **Servo Systems**

Precision 5.0V FGA™ Voltage Reference

X60008C-50
X60008D-50

ELECTRICAL CHARACTERISTICS

(Operating Conditions: $V_{IN} = 6.5V$, $I_{OUT} = 0mA$, $C_{OUT} = 0.001\mu F$, $T_A = -40$ to $+85^\circ C$ unless otherwise specified.)

Symbol	Parameter	Conditions	Min	Typ	Max	Units
V_{OUT}	Output Voltage			5.000		V
V_{OA}	V_{OUT} Accuracy X60008CIS8-50 X60008DIS8-50	$T_A = 25^\circ C$	-0.50 -1.00		+0.50 +1.00	mV
I_{IN}	Supply Current			500	800	nA
V_{IN}	Input Voltage Range		5.1		9.0	V
TC V_{OUT}	Output Voltage Temperature Coefficient ⁽¹⁾	X60008CIS8-50 X60008DIS8-50			5 10	ppm/ $^\circ C$
$\Delta V_{OUT}/\Delta V_{IN}$	Line Regulation	$+5.5V \leq V_{IN} \leq +8.0V$			100	$\mu V/V$
$\Delta V_{OUT}/\Delta I_{OUT}$	Load Regulation	$0mA \leq I_{SOURCE} \leq 10mA$ $-10mA \leq I_{SINK} \leq 0mA$		15 25	50 100	$\mu V/mA$
$\Delta V_{OUT}/\Delta t$	Long Term Stability	$T_A = 25^\circ C$		10		ppm/ 1000Hrs
$\Delta V_{OUT}/\Delta T_A$	Thermal Hysteresis ⁽²⁾	$\Delta T = -40^\circ C$ to $+85^\circ C$		50		ppm
V_{DO}	Dropout Voltage ⁽³⁾	$I_{OUT} = 5mA$, $\Delta V_{OUT} = -0.01\%$		150	300	mV
I_{SC}	Short Circuit Current ⁽⁴⁾	$T_A = 25^\circ C$		50	80	mA
V_N	Output Voltage Noise	0.1Hz to 10Hz		30		μV_{pp}

- Note:**
- Over the specified temperature range. Temperature coefficient is measured by the box method whereby the change in V_{OUT} is divided by the temperature range; in this case, $-40^\circ C$ to $+85^\circ C = 125^\circ C$.
 - Thermal Hysteresis is the change in V_{OUT} created by package stress @ $T_A = 25^\circ C$ after temperature cycling. V_{OUT} is read initially at $T_A = 25^\circ C$; the X60008 is then cycled between Hot ($85^\circ C$) and Cold ($-40^\circ C$) before a second V_{OUT} measurement is taken at $25^\circ C$. The deviation between the initial V_{OUT} reading and the second V_{OUT} reading is then expressed in ppm.
 - Dropout voltage (V_{DO}) is the minimum voltage (V_{IN}) into the X60008 which will produce the output voltage (ΔV_{OUT}) drop specified in the Electrical Characteristics table.
 - Guaranteed by Device Characterization

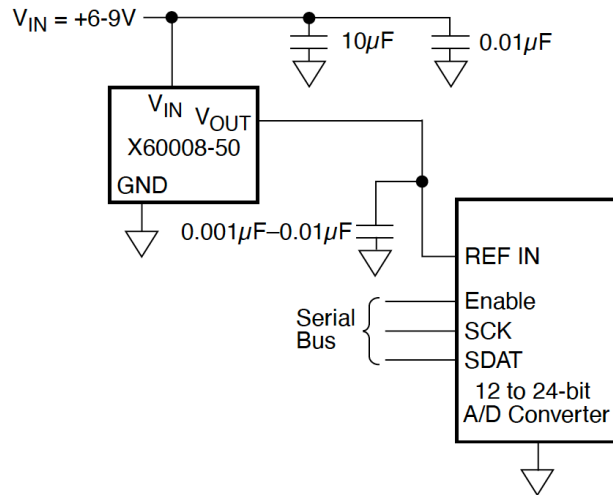
Ejemplo



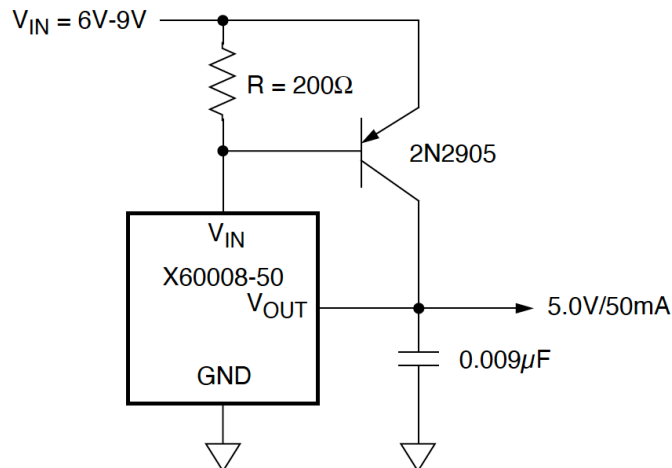
Precision 5.0V FGA™ Voltage Reference

X60008C-50
X60008D-50

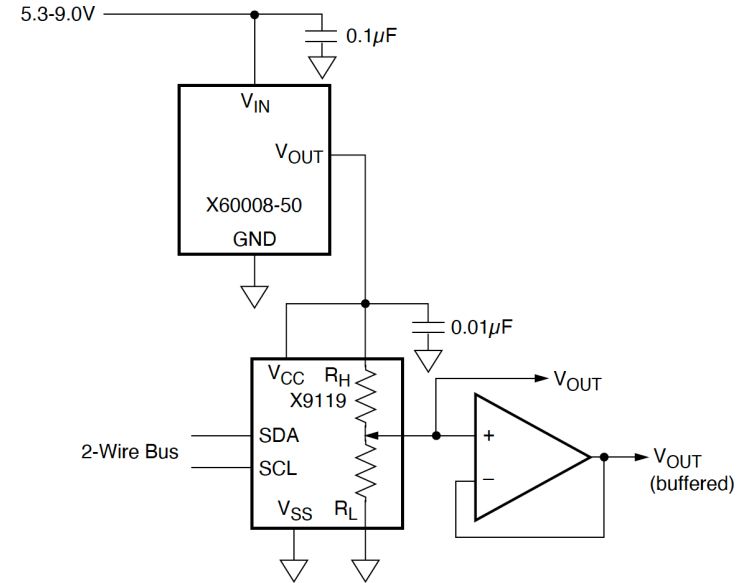
TYPICAL APPLICATION CIRCUITS



Precision 5V, 50mA Reference.



5V Full Scale Low-Drift 10-bit Adjustable Voltage Source



±5.0V Dual Output, High Accuracy Reference

